

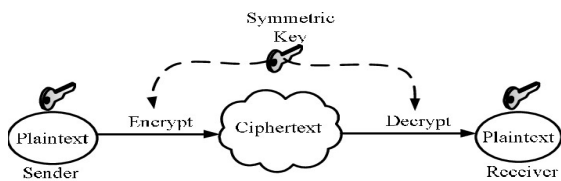
UNIT-I

Q1. Illustrate any five of the following terms with a neat sketch

(a) Confidentiality

Confidentiality ensures that information is **accessible only to authorized users** and is **not disclosed to unauthorized persons**.

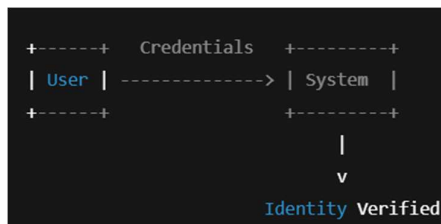
Example: Encryption of bank account details during online transactions.



(b) Authentication

Authentication is the process of **verifying the identity of a user or system** before allowing access.

Example: Login using username–password or OTP verification.



(c) Availability

Availability ensures that **systems, services, and data are accessible** to authorized users whenever required.

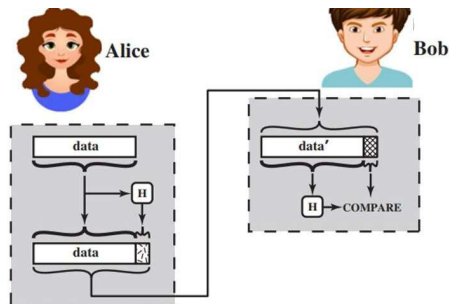
Example: Banking servers providing service 24×7.



(d) Integrity

Integrity ensures that **data remains accurate, complete, and unaltered** during transmission or storage.

Example: Use of hash functions to detect data modification.

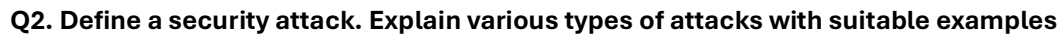


(e) Access Control

Example: File permissions such as read, write, and execute.



Example: Digital signatures in online transactions.



Types of Security Attacks

1. Passive Attacks

(a) Eavesdropping / Snooping

Example: Attacker capturing passwords over an unsecured Wi-Fi network.

(b) Traffic Analysis

Example: Identifying active users by analyzing network traffic volume.

2. Active Attacks: Active attacks involve **modifying, deleting, or disrupting** data or services.

(a) Masquerade

Example: Using stolen login credentials to access a system.

(b) Replay Attack

Valid data is captured and retransmitted later by the attacker.

Example: Reusing a captured authentication message to gain access.

(c) Modification of Messages

Data is altered during transmission.

Example: Changing transaction amount in online banking.

(d) Denial of Service (DoS)

System resources are flooded to make services unavailable.

Example: Flooding a server with requests to crash it.

3. Specific Attack Techniques:

Malware: Malicious software (viruses, worms, ransomware) designed to damage or infiltrate systems.

Social Engineering (Phishing): Tricking individuals into revealing sensitive information.

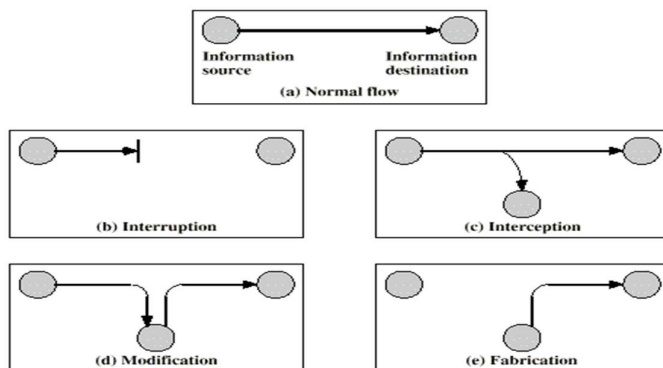
Man-in-the-Middle (MITM): Intercepting communication between two parties.

SQL Injection: Inserting malicious code into web applications to manipulate databases.

Q3. Define a security attack and explain the concepts of interruption, interception, modification, and fabrication

A **security attack** is any action that **attempts to compromise the security of information** by violating confidentiality, integrity, or availability.

Types of Security Attacks



1. Interruption

Interruption occurs when **system resources or data become unavailable** to authorized users.

- **Security goal affected:** Availability
- **Example:** Server crash or Denial of Service (DoS) attack

Sender —X—► Receiver

(System or data unavailable)

2. Interception

Interception occurs when **unauthorized users gain access to data**.

- **Security goal affected:** Confidentiality

- **Example:** Eavesdropping on network communication

Sender —► Attacker —► Receiver

(Data is read by attacker)

3. Modification

Modification occurs when **data is altered** during transmission.

- **Security goal affected:** Integrity
- **Example:** Changing the amount in an online banking transaction

Sender —► Altered Data —► Receiver

4. Fabrication

Fabrication occurs when **false data or messages are injected** into the system.

- **Security goal affected:** Integrity & Authenticity
- **Example:** Fake emails or forged transactions

Attacker —► Fake Message —► Receiver

Interruption → Availability

Interception → Confidentiality

Modification → Integrity

Fabrication → Authenticity

Q4(a). Define the three CIA security goals with one real-world example each

1. Confidentiality: Confidentiality ensures that **information is accessible only to authorized users.**

Real-world example:

In online banking systems, sensitive information such as **ATM PINs, passwords, and account numbers** is encrypted before transmission. Even if an attacker intercepts the data on the network, the information appears as unreadable ciphertext and cannot be understood without the decryption key, thus maintaining confidentiality.

2. Integrity: Integrity ensures that **data remains accurate and unaltered** during storage or transmission.

Real-world example:

When software or files are downloaded from a website, a **hash value** is generated for the original file. After download, the receiver computes the hash again and compares it with the original. If the values match, it confirms that the file has not been modified by an attacker during transmission.

3. Availability: Availability ensures that **systems and services are available whenever needed** by authorized users.

Real-world example:

Banking servers use **backup servers, load balancing, and redundant network connections** to ensure continuous service. Even if one server fails or is under attack, another server takes over, allowing customers to access banking services 24×7 without downtime.



Q4(b). Differentiate between passive and active attacks with examples

Passive Attacks

Passive attacks involve **monitoring or listening to communication** without affecting the system resources or data.

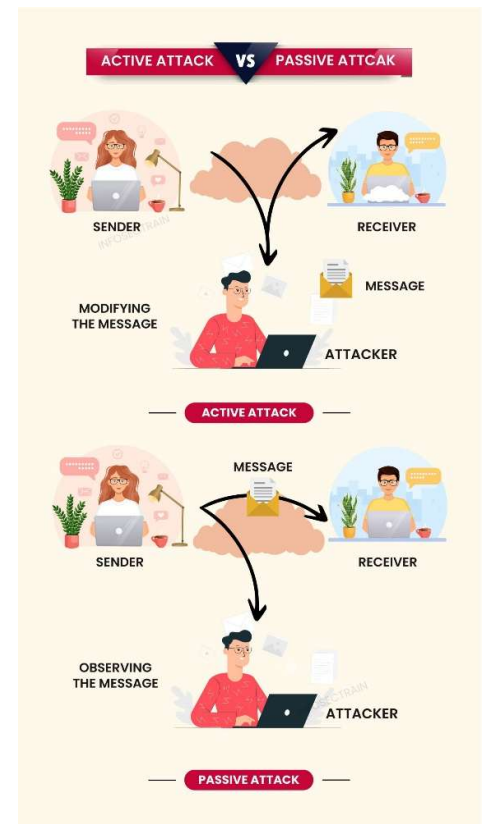
- **Security goal affected:** Confidentiality
- **Characteristics:** Difficult to detect, no data modification
- **Examples:** Eavesdropping, Traffic analysis

Active Attacks

Active attacks involve **altering, deleting, or disrupting data or services**.

- **Security goals affected:** Integrity and Availability
- **Characteristics:** Can be detected, causes system damage
- **Examples:**
 - Message modification
 - Denial of Service (DoS) attack

Aspect	Passive Attack	Active Attack
Nature	Observes data	Alters or disrupts data
Data change	No	Yes
Detection	Difficult	Easier
Impact	Confidentiality	Integrity & Availability



Q5. Given snooping and modification attacks, recommend TWO security services and TWO security mechanisms. Explain the mapping of mechanism to the services.

Snooping and modification are common network security attacks that threaten the confidentiality and integrity of data. To protect against these attacks, suitable **security services** are provided using appropriate **security mechanisms**.

1. Snooping Attack: Snooping is a **passive attack** in which an attacker secretly **monitors or reads data** during transmission without altering it.

Example

An attacker capturing usernames and passwords over an unsecured Wi-Fi network.

Security Goal Affected: Confidentiality

2. Modification Attack: Modification is an **active attack** in which data is **altered during transmission**.

Example

Changing the amount in an online banking transaction while the data is in transit.

Security Goals Affected: Integrity, Authentication

Recommended Security Services

Service 1: Confidentiality

Confidentiality service protects information from unauthorized disclosure and prevents snooping attacks.

Service 2: Integrity

Integrity service ensures that data is not altered during transmission, thus preventing modification attacks.

Recommended Security Mechanisms

Mechanism 1: Encryption

Encryption converts plaintext into ciphertext using a secret key, making the data unreadable to unauthorized users.

Mechanism 2: Hash Function / Message Authentication Code (MAC)

Hash functions or MACs generate a fixed-length value used to detect any change in data.

Mapping of Security Mechanisms to Security Services

Security Service	Security Mechanism	Explanation
Confidentiality	Encryption	Encrypted data cannot be understood by snoopers
Integrity	Hash / MAC	Any modification changes the hash value
Confidentiality	Encryption	Prevents unauthorized data access
Integrity	MAC / Digital Signature	Detects message alteration and verifies authenticity

Q6. Alice sends confidential bank data. Classify possible attacks (passive/active) and suggest countermeasures.

When Alice sends confidential bank data over a network, the data may be exposed to various **security attacks**. These attacks can be broadly classified into **passive** and **active** attacks. Appropriate **countermeasures** are required to protect the data.

1. Passive Attacks

Passive attacks involve **monitoring or listening** to the communication without modifying the data.

(a) Eavesdropping / Snooping : An attacker secretly reads Alice's bank data during transmission.

Effect: Leads to loss of **confidentiality**.

(b) Traffic Analysis : The attacker observes communication patterns to infer sensitive information.

Effect: May reveal banking activity details.

Countermeasures for Passive Attacks

- **Encryption** of bank data before transmission
- Use of **secure communication protocols** (HTTPS, SSL/TLS)

2. Active Attacks: Active attacks involve **altering, deleting, or injecting data**.

(a) Modification of Data: Attacker changes transaction details such as account number or amount.

Effect: Loss of **integrity**.

(b) Replay Attack : Attacker captures a valid transaction and retransmits it later.

Effect: Unauthorized transactions.

(c) Masquerade: Attacker pretends to be Alice using stolen credentials.

Effect: Loss of **authentication and integrity**.

Countermeasures for Active Attacks

- **Message Authentication Codes (MACs) or digital signatures**
- **Timestamping and nonces** to prevent replay attacks
- **Strong authentication mechanisms** (OTP, multi-factor authentication)

Q7. Classify attacks as passive and active, and describe any two passive and two active attacks relevant to networked systems.

In network security, a **security attack** is any action that compromises the confidentiality, integrity, or availability of information. Based on the behavior of the attacker, attacks are broadly classified into **passive attacks** and **active attacks**.

Classification of Security Attacks

Passive Attacks :Passive attacks involve **monitoring or observing communication** without altering the data or system resources. These attacks are difficult to detect because no changes are made to the data.

Active Attacks: Active attacks involve **modification, deletion, or disruption** of data or network services. These attacks can be detected but cause serious damage.

Passive Attacks (Any Two)

1. Eavesdropping (Snooping) : Eavesdropping is a passive attack in which an attacker secretly listens to data transmission between sender and receiver.

Example: An attacker captures sensitive information such as usernames, passwords, or bank details over an unsecured wireless network.

Impact:

- Violates confidentiality
- Leads to identity theft and financial loss

Detection: Difficult to detect as data is not altered.

2. Traffic Analysis: Traffic analysis is a passive attack where the attacker studies **communication patterns**, even if the data is encrypted.

Example: An attacker analyzes network traffic to identify peak transaction times in a banking system.

Impact:

- Reveals communication behavior
- Can be used to plan future attacks

Detection: Very difficult due to lack of data modification.

Active Attacks (Any Two)

1. Modification of Messages: In this attack, the attacker **alters the content of messages** during transmission.

Example: Changing the transfer amount or beneficiary account number in online banking.

Impact:

- Violates data integrity
- Causes financial and operational damage

Detection: Possible using integrity checks such as MACs or hashes.

2. Denial of Service (DoS): A Denial of Service attack floods the target system with excessive traffic, making it unavailable to legitimate users.

Example: Overloading an e-commerce website with requests to crash the server.

Impact:

- Violates availability
- Causes service downtime and revenue loss

Detection: Possible through abnormal traffic monitoring.

Feature	Passive Attacks	Active Attacks
Nature	Observation only	Modification / Disruption
Data alteration	No	Yes
Detection	Very difficult	Easier
Security goals affected	Confidentiality	Integrity, Availability

Q8. Solve the following:

(i) Compute $\gcd(252, 198)$ using the Euclidean algorithm and express it as $sa + tb$

Step 1: Apply Euclidean algorithm

$$\begin{aligned}252 &= 198 \times 1 + 54 \\198 &= 54 \times 3 + 36 \\54 &= 36 \times 1 + 18 \\36 &= 18 \times 2 + 0 \\ \therefore \gcd(252, 198) &= 18\end{aligned}$$

Step 2: Express gcd as linear combination

$$\begin{aligned}18 &= 54 - 36 \\36 &= 198 - 54 \times 3 \\18 &= 54 - (198 - 54 \times 3) \\18 &= 4(54) - 198 \\54 &= 252 - 198 \\18 &= 4(252 - 198) - 198 \\18 &= 4(252) - 5(198) \\S=4, t=-5\end{aligned}$$

(ii) Solve $7x \equiv 3 \pmod{26}$ and interpret the result in terms of decryption of a shift cipher

Step 1: Find modular inverse of 7 mod 26

We find x such that:

$$\begin{aligned}7x &\equiv 1 \pmod{26} \\7 \times 15 &= 105 \equiv 1 \pmod{26}\end{aligned}$$

So, inverse of 7 mod 26 is **15**.

Step 2: Multiply both sides by inverse

$$\begin{aligned}x &\equiv 15 \times 3 \pmod{26} \\x &\equiv 45 \equiv 19 \pmod{26}\end{aligned}$$

$$\boxed{x = 19}$$

Interpretation (Shift Cipher): This result means that a **shift of 19** positions is used in decryption. Each encrypted letter is shifted **19 positions backward** to obtain the original plaintext.

Q9. Design a simple encryption method using modular arithmetic and demonstrate how both encryption and decryption work.

A simple encryption method can be designed using **modular arithmetic**, where letters are represented by numbers and mathematical operations are applied to encrypt and decrypt the message.

Plaintext Representation

Let each letter be represented as a number:

Letter	A	B	C	...	Z
Value	0	1	2	...	25

Encryption Method

Let:

- Plaintext letter = **P**
- Ciphertext letter = **C**
- Key = **k = 5**

Encryption formula: $C = (P + k) \bmod 26$

Encryption Example

Plaintext: **HELLO**

Letter	H	E	L	L	O
P	7	4	11	11	14
$C = (P+5) \bmod 26$	12	9	16	16	19
Ciphertext	M	J	Q	Q	T

Ciphertext: MJQQT

Decryption Method

Decryption formula: $P = (C - k) \bmod 26$

Decryption Example

Ciphertext: **MJQQT**

Letter	M	J	Q	Q	T
C	12	9	16	16	19
$P = (C-5) \bmod 26$	7	4	11	11	14
Plaintext	H	E	L	L	O

Recovered Plaintext: HELLO

This simple encryption method uses **modular arithmetic** to securely transform plaintext into ciphertext and back using a secret key.

Q10. Evaluate which security goal—confidentiality, integrity, or availability—is most impacted by a replay attack, with justification.

A **replay attack** is a type of security attack in which an attacker **captures valid data transmissions** and **retransmits them later** to produce an unauthorized effect. Replay attacks are common in networked and authentication-based systems.

Security Goals Overview

- **Confidentiality:** Prevents unauthorized disclosure of information
- **Integrity:** Prevents unauthorized modification of data
- **Availability:** Ensures systems and services are accessible when required

Most Impacted Security Goal: Integrity

The security goal **most impacted by a replay attack is *Integrity***.

Justification

1. In a replay attack, the attacker **does not change the content** of the message.
→ Therefore, **confidentiality is not directly violated**.
2. The attacker **reuses a valid, previously transmitted message** and presents it as a new one.
→ This causes the system to **accept false or duplicate transactions**.
3. As a result, the system processes **unauthorized actions**, such as repeated fund transfers.
→ This violates the **correctness and trustworthiness of operations**, i.e., integrity.

Example

In online banking, an attacker captures a valid transaction request and replays it later. Although the message is unchanged and readable only by authorized users, the system executes the transaction again, causing **duplicate or fraudulent transfers**.

Secondary Impact

- **Availability** may also be affected if replayed messages overload the system.
- However, the **primary violation remains integrity**.

Prevention Measures

- Use of **timestamps**
- Use of **nonces**
- Sequence numbers in communication protocols

UNIT 2

Q1. With a neat diagram explain the DES algorithm

The **Data Encryption Standard (DES)** is a **symmetric key block cipher** that encrypts data in **64-bit blocks** using a **56-bit secret key**. It is based on the **Feistel network structure** and performs **16 rounds** of encryption to produce ciphertext.

Basic Features of DES

- Block size: **64 bits**
- Key size: **56 bits** (8 bits used for parity)
- Number of rounds: **16**
- Type: **Symmetric key encryption**
- Structure: **Feistel cipher**

Working of DES Algorithm

The DES algorithm consists of the following steps:

1. Initial Permutation (IP)

- The 64-bit plaintext is rearranged using a fixed **initial permutation table**.
- This step increases diffusion by spreading bits across positions.

2. Splitting the Plaintext

- The permuted plaintext is divided into:
 - Left half **L_0 (32 bits)**
 - Right half **R_0 (32 bits)**

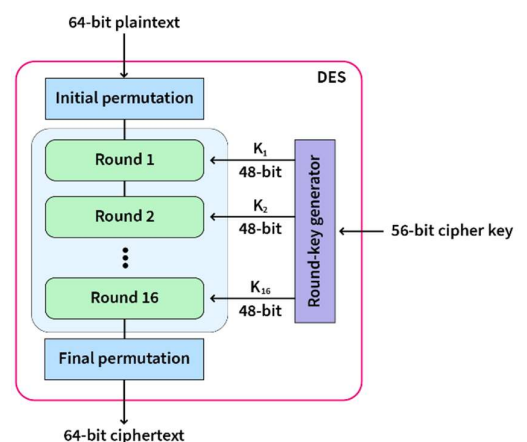
3. Key Generation

- The original 64-bit key is reduced to **56 bits** by removing parity bits.
- The key is split into two **28-bit halves**.
- For each round:
 - Left circular shifts are applied
 - A **48-bit round key** is generated
- Total **16 round keys (K_1 to K_{16})** are produced.

4. DES Round Function (16 Rounds)

Each round consists of the following operations:

- **Expansion:**
Expands 32-bit R_{i-1} to **48 bits**
- **Key Mixing:**
XORs expanded data with round key K_i
- **Substitution:**
Uses **8 S-boxes** to compress data back to 32 bits (confusion)



- **Permutation:**
Rearranges bits for diffusion
- **XOR and Swap:**
Output is XORed with L_{i-1} and halves are swapped

5. Final Permutation (FP)

- After the 16th round, the swap is undone
- The inverse of the initial permutation is applied
- Produces the **64-bit ciphertext**

Decryption in DES

- Decryption uses the **same algorithm**
- The only difference is that **round keys are applied in reverse order**
- This is possible due to the Feistel structure

Advantages of DES

- Simple and efficient in hardware
- Well-studied and standardized
- Same process for encryption and decryption

Limitations of DES

- Small key size (56 bits)
- Vulnerable to brute-force attacks
- Not secure for modern applications

Q2(a). Discuss the strengths of DES algorithm

1. Well-studied algorithm

DES has been extensively analyzed for many years, and its strengths and weaknesses are well understood.

2. Strong confusion and diffusion

The use of S-boxes and permutations provides good confusion and diffusion, making statistical attacks difficult.

3. Feistel structure

The Feistel network allows the same algorithm to be used for both encryption and decryption, simplifying implementation.

4. Efficient hardware implementation

DES is fast and efficient when implemented in hardware.

5. Standardized algorithm

DES was adopted as a federal standard, which led to wide acceptance and interoperability.

Q2(b). Discuss about 3DES algorithm with a neat sketch

Triple DES (3DES) is an enhancement of DES designed to overcome its small key size. It applies the DES algorithm **three times** on the same data using **multiple keys**, thereby increasing security.

Working of 3DES

- 3DES uses **three DES operations**:
 1. **Encryption with Key K_1**
 2. **Decryption with Key K_2**
 3. **Encryption with Key K_3**

This is known as **EDE mode (Encrypt–Decrypt–Encrypt)**.

Keying Options

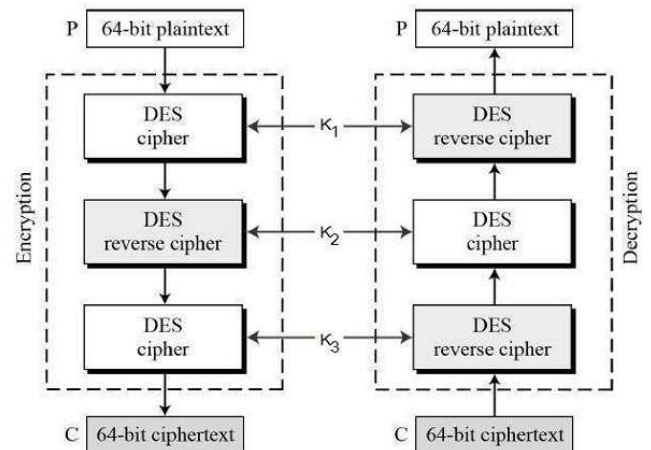
- **Three keys (K_1, K_2, K_3)**: Most secure (168-bit key)
- **Two keys ($K_1 = K_3$)**: Effective key size of 112 bits
- **One key ($K_1 = K_2 = K_3$)**: Equivalent to DES

Advantages of 3DES

- Much more secure than DES
- Backward compatible with DES
- Resistant to brute-force attacks

Limitations of 3DES

- Slower than DES and AES
- Being phased out in favor of AES



Q3. Discuss about the AES algorithm with a neat sketch

The **Advanced Encryption Standard (AES)** is a **symmetric key block cipher** adopted as a standard to replace DES. It is widely used due to its **high security, efficiency, and flexibility**. AES operates on a **128-bit data block** and supports key sizes of **128, 192, and 256 bits**.

Basic Features of AES

- Block size: **128 bits**
- Key sizes: **128 / 192 / 256 bits**
- Number of rounds:
 - 10 rounds (128-bit key)
 - 12 rounds (192-bit key)
 - 14 rounds (256-bit key)
- Structure: **Substitution–Permutation Network (SPN)**
- Type: **Symmetric key encryption**

Overall Working of AES

AES encryption consists of an **initial round**, followed by **multiple main rounds**, and a **final round**.

1. State Matrix Formation

- The 128-bit plaintext is arranged into a **4 × 4 byte matrix** called the **state matrix**.
- All AES operations are performed on this state matrix.

2. Key Expansion

- The original cipher key is expanded to generate **round keys**.
- Each round uses a different **128-bit round key**.

3. Initial Round

- **AddRoundKey** operation is applied.
- The plaintext state is XORed with the first round key.

4. Main Rounds (Repeated Rounds)

Each round consists of the following four transformations:

1. **SubBytes**
 - Each byte is replaced using an S-box.
 - Provides **confusion**.
2. **ShiftRows**
 - Rows of the state matrix are cyclically shifted.
 - Provides **diffusion**.
3. **MixColumns**
 - Columns are mixed using matrix multiplication in $GF(2^8)$.
 - Further increases diffusion.
4. **AddRoundKey**
 - State matrix is XORed with the round key.

5. Final Round

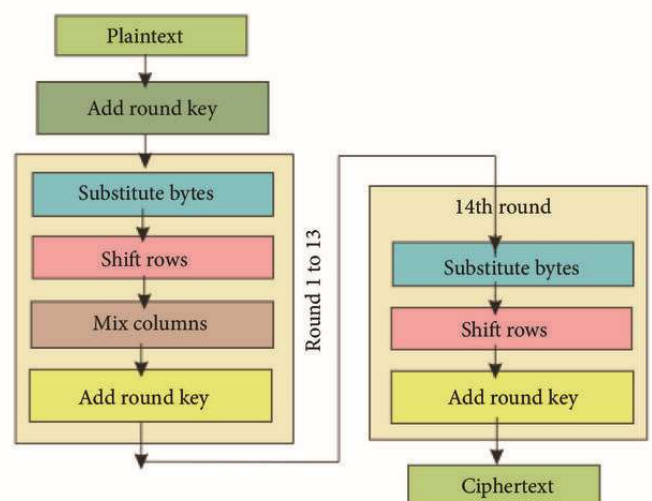
- Same as main rounds but **without MixColumns**.
- Produces the final **ciphertext**.

AES Decryption

- Uses **inverse transformations**:
 - InvSubBytes
 - InvShiftRows
 - InvMixColumns
 - AddRoundKey
- Round keys are applied in **reverse order**.

Advantages of AES

- Strong security with large key sizes
- Resistant to known cryptographic attacks
- Efficient in both hardware and software
- Widely used in real-world applications (SSL/TLS, disk encryption)



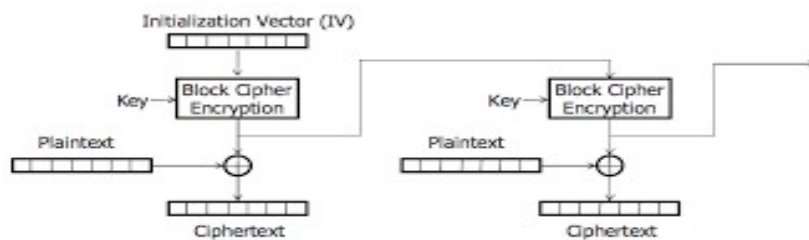
Q4. List the difference between block cipher and stream cipher. Explain block cipher modes of operation.

Difference between Block Cipher and Stream Cipher

Aspect	Block Cipher	Stream Cipher
Data processing	Encrypts data in fixed-size blocks	Encrypts data bit by bit or byte by byte
Block size	Fixed (e.g., 64-bit, 128-bit)	No fixed block size
Speed	Slower than stream ciphers	Faster
Error effect	Error affects entire block	Error affects only one bit/byte
Examples	DES, AES	RC4

Block Cipher Modes of Operation

Block cipher modes define **how blocks are encrypted** and chained together to provide better security.



1. Electronic Code Book (ECB) Mode

- Each plaintext block is encrypted **independently**.
- Same plaintext blocks produce same ciphertext blocks.

Advantages: Simple and fast

Disadvantages:

- Not secure for sensitive data
- Patterns are visible

2. Cipher Block Chaining (CBC) Mode

- Each plaintext block is XORed with the **previous ciphertext block** before encryption.
- Uses an **Initialization Vector (IV)**.

Advantages: Hides patterns, More secure than ECB

Disadvantages: Error in one block affects next block

3. Cipher Feedback (CFB) Mode

- Converts block cipher into a **stream cipher**.
- Uses feedback from previous ciphertext.

Advantages: Suitable for real-time data

Disadvantages: Slower than ECB

4. Output Feedback (OFB) Mode

- Uses output of encryption function as feedback.

- No error propagation.

Advantages: Errors do not spread

Disadvantages: Requires synchronization

5. Counter (CTR) Mode

- Uses a counter value combined with a key.
- Same plaintext encrypts to different ciphertexts.

Advantages: Fast, Supports parallel processing

Disadvantages: Counter reuse can break security

Q5. Compare DES and AES with respect to key size, block size, security level, and design approach.

DES and AES are **symmetric key block ciphers** used for data encryption. DES is an older standard, while AES is a modern encryption standard designed to overcome the limitations of DES.

Comparison between DES and AES

Feature	DES (Data Encryption Standard)	AES (Advanced Encryption Standard)
Key size	56 bits	128, 192, or 256 bits
Block size	64 bits	128 bits
Number of rounds	16 rounds	10, 12, or 14 rounds (based on key size)
Security level	Low (vulnerable to brute-force attacks)	High (resistant to known attacks)
Design approach	Feistel structure	Substitution–Permutation Network (SPN)
Speed & efficiency	Efficient in hardware, slower in software	Efficient in both hardware and software
Flexibility	Fixed key and block size	Flexible key sizes
Current usage	Obsolete / insecure	Widely used standard

Security Comparison

- DES has a **small key size**, making it susceptible to brute-force attacks.
- AES uses **larger key sizes**, providing strong resistance against cryptographic attacks.

Design Approach Difference

- **DES** uses a **Feistel network**, where the same structure is used for encryption and decryption.
- **AES** uses a **Substitution–Permutation Network**, providing better diffusion and confusion

Q6. Describe the four main transformations used in AES (SubBytes, ShiftRows, MixColumns, AddRoundKey).

AES encryption is based on a series of **round transformations** applied to the **state matrix**. Each round consists of four main transformations that provide **confusion and diffusion**, ensuring strong security.

1. SubBytes Transformation

- SubBytes is a **non-linear byte substitution** step.
- Each byte in the state matrix is replaced using a **substitution box (S-box)**.
- The S-box is designed to resist linear and differential cryptanalysis.

Purpose: Provides **confusion** by obscuring the relationship between plaintext and ciphertext.

2. ShiftRows Transformation

- ShiftRows is a **row-wise permutation** operation.
- Rows of the state matrix are cyclically shifted to the left:
 - Row 0: No shift
 - Row 1: 1-byte shift
 - Row 2: 2-byte shift
 - Row 3: 3-byte shift

Purpose: Spreads the influence of each byte across columns, providing **diffusion**.

3. MixColumns Transformation

- MixColumns treats each column as a polynomial over $\text{GF}(2^8)$.
- Columns are multiplied with a **fixed polynomial matrix**.
- This operation mixes the bytes within each column.

Purpose: Further enhances **diffusion**, making relationships between input and output complex.

4. AddRoundKey Transformation

- The state matrix is XORed with the **round key**.
- Round keys are derived from the original cipher key using key expansion.

Purpose: Introduces **key dependency**, ensuring that encryption depends on the secret key.

Order of Transformations in AES

- **Initial Round:** AddRoundKey
- **Main Rounds:** SubBytes → ShiftRows → MixColumns → AddRoundKey
- **Final Round:** SubBytes → ShiftRows → AddRoundKey (no MixColumns)

Q7(a). Interpret the purpose of algebraic structures in symmetric key cryptography

Algebraic structures such as **groups, rings, and finite fields** play a crucial role in symmetric key cryptography by providing a **mathematical foundation** for encryption and decryption operations.

Purpose and importance:

1. **Well-defined operations**
Algebraic structures ensure that operations like addition, multiplication, and inversion are mathematically well-defined and predictable.
2. **Security through complexity**
Operations in finite fields (e.g., $\text{GF}(2^n)$) make it computationally difficult for attackers to reverse encryption without the key.
3. **Confusion and diffusion**
Algebraic transformations help achieve confusion and diffusion, which are essential properties for secure ciphers.
4. **Key-dependent transformations**
Mathematical structures allow encryption operations to depend strongly on the secret key.

5. Efficient implementation

Algebraic operations can be efficiently implemented in both hardware and software.

Example:

AES uses arithmetic in $\text{GF}(2^8)$ for SubBytes and MixColumns operations.

Q7(b). Compute the result of addition in $\text{GF}(2^4)$ for two given polynomials and apply it to a cipher operation

Addition in $\text{GF}(2^4)$

In $\text{GF}(2^4)$, addition is performed using **bitwise XOR**, and **no carry** is generated.

Let the two polynomials be:

$$\begin{aligned}A(x) &= x^3 + x + 1(1011) \\B(x) &= x^3 + x^2 + 1(1101)\end{aligned}$$

Step 1: Perform XOR operation

$$1011 \oplus 1101 = 0110$$

Step 2: Write the result as a polynomial

$$0110 = x^2 + x$$

Application in Cipher Operation

- Such addition is used in **key mixing operations**, where plaintext/state bytes are combined with key bytes.
- Since XOR is reversible, the same operation can be used during decryption.

Q8(a). Show how to multiply (10101) by (10000) in $\text{GF}(2^5)$. Use $(x^5 + x^2 + 1)$ as modulus

Step 1: Represent the binary numbers as polynomials

$$\begin{aligned}(10101)_2 &= x^4 + x^2 + 1 \\(10000)_2 &= x^4\end{aligned}$$

Step 2: Multiply the polynomials

$$(x^4 + x^2 + 1)(x^4) = x^8 + x^6 + x^4$$

Step 3: Reduce modulo $x^5 + x^2 + 1$

Since:

$$x^5 \equiv x^2 + 1$$

Now reduce higher powers:

$$\begin{aligned}x^8 &= x^3(x^5) = x^3(x^2 + 1) = x^5 + x^3 \\x^5 &= x^2 + 1\end{aligned}$$

So:

$$\begin{aligned}x^8 &= (x^2 + 1) + x^3 = x^3 + x^2 + 1 \\x^6 &= x(x^5) = x(x^2 + 1) = x^3 + x\end{aligned}$$

Step 4: Substitute back

$$\begin{aligned}&x^8 + x^6 + x^4 \\&= (x^3 + x^2 + 1) + (x^3 + x) + x^4\end{aligned}$$

Step 5: Combine like terms (mod 2)

$$= x^4 + x^2 + x + 1$$

Final Answer

$$x^4 + x^2 + x + 1 = (10111)_2$$

Q8(b). Outline why $\text{GF}(2^n)$ fields are used in modern symmetric key algorithms

Finite fields of the form $\text{GF}(2^n)$ are widely used in modern symmetric key algorithms due to the following reasons:

1. **Binary compatibility**
Elements of $\text{GF}(2^n)$ map naturally to binary numbers, making them ideal for digital systems.
2. **Simple operations**
Addition is performed using XOR, which is fast and efficient in hardware and software.
3. **Strong mathematical structure**
Every non-zero element has a multiplicative inverse, enabling secure encryption transformations.
4. **Security strength**
Algebraic complexity in $\text{GF}(2^n)$ provides resistance against cryptographic attacks.
5. **Practical usage**
Modern algorithms like AES use $\text{GF}(2^8)$ for SubBytes and MixColumns operations.

Q9(a). Define a field. Why $\mathbb{Z}_5$ is a field but $\mathbb{Z}_6$ is not?

A **field** is an algebraic structure consisting of a set of elements with two operations, **addition (+)** and **multiplication ($\times$)**, such that:

1. The set is **closed** under addition and multiplication
2. There exist **additive and multiplicative identities**
3. Every element has an **additive inverse**
4. Every non-zero element has a **multiplicative inverse**
5. Both operations are **associative and commutative**
6. Multiplication is **distributive over addition**

Why $\mathbb{Z}_5$ is a Field

$$\mathbb{Z}_5 = \{0, 1, 2, 3, 4\}$$

- All operations are performed **modulo 5**
- Since 5 is a **prime number**, every non-zero element has a **multiplicative inverse**

Examples:

- Inverse of 2 is 3 ($2 \times 3 \equiv 1 \pmod{5}$)
- Inverse of 4 is 4 ($4 \times 4 \equiv 1 \pmod{5}$)

Hence, $\mathbb{Z}_5$ satisfies all field properties and is a field.

Why $\mathbb{Z}_6$ is NOT a Field

$$\mathbb{Z}_6 = \{0, 1, 2, 3, 4, 5\}$$

- Operations are modulo 6
- 6 is **not a prime number**
- Some non-zero elements **do not have multiplicative inverses**

Example:

- 2 has no inverse mod 6 because
 $2 \times \text{any element} \neq 1 \pmod{6}$

Hence, $\mathbb{Z}_6$ is **not a field**.

Q9(b). Discuss the properties of Group and Cyclic Group algebraic structures with examples

Group

A **group** is a set of elements with a single operation ($\star$) that satisfies:

1. **Closure** – $a \star b$ is in the set
2. **Associativity** – $(a \star b) \star c = a \star (b \star c)$
3. **Identity element** – $e \star a = a$
4. **Inverse element** – $a \star a^{-1} = e$

Example: Integers under addition ($\mathbb{Z}, +$) form a group.

Cyclic Group

A **cyclic group** is a group that can be generated using a **single element**, called a **generator**.

Every element of the group can be written as powers (or multiples) of the generator.

Example: $\mathbb{Z}_6$ under addition is a cyclic group generated by 1.

Difference Between Group and Cyclic Group

Aspect	Group	Cyclic Group
Generator	Not necessary	Single generator
Structure	General	Special type of group
Example	$\mathbb{Z}$ under +	$\mathbb{Z}_6$ under +

Application in Cryptography

- Groups and cyclic groups form the basis of **modular arithmetic**
- Used in encryption, key generation, and cryptographic transformations

Q10. Compare the design differences between modern block ciphers and classical block ciphers.

Block ciphers are encryption algorithms that operate on fixed-size blocks of data. Based on their development period and design philosophy, block ciphers are classified into **classical block ciphers** and **modern block ciphers**. They differ significantly in structure, security, and efficiency.

Classical Block Ciphers: Classical block ciphers were designed in the early stages of cryptography and are relatively simple.

Characteristics

- Based mainly on **substitution and transposition**
- Small key sizes
- Limited number of rounds
- Simple mathematical operations
- Vulnerable to brute-force and cryptanalysis attacks

Examples: DES (to some extent), Early substitution–permutation ciphers

Modern Block Ciphers: Modern block ciphers are designed using strong mathematical foundations and advanced cryptographic principles.

Characteristics

- Use complex algebraic structures such as **finite fields**
- Large key sizes
- Multiple rounds for strong security
- Provide strong **confusion and diffusion**
- Resistant to known cryptographic attacks

Examples: AES, 3DES

Aspect	Classical Block Ciphers	Modern Block Ciphers
Key size	Small	Large
Block size	Small / fixed	Larger (e.g., 128-bit)
Number of rounds	Few	Many
Mathematical basis	Simple substitutions	Finite fields, algebra
Security strength	Low	High
Resistance to attacks	Weak	Strong
Implementation	Simple	Optimized for HW & SW

Security Comparison

- Classical block ciphers are vulnerable to **frequency analysis, brute-force attacks**, and other cryptanalytic methods.
- Modern block ciphers are designed to resist **linear, differential, and brute-force attacks**.